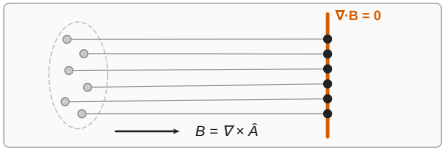
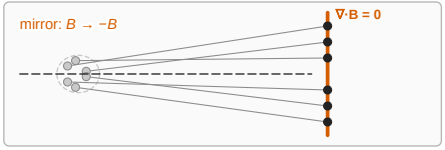
Model design

W4 Vector potentials



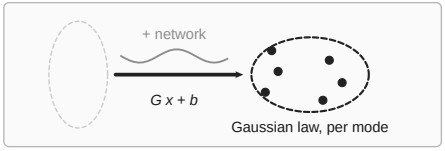
the output lies on the surface by construction

W4s B → −B equivariance



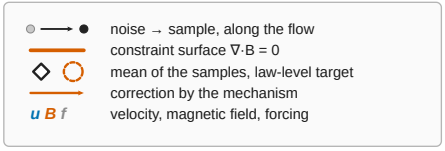
the velocity field is equivariant

W9, W10, W11 Affine term, Gaussian anchor



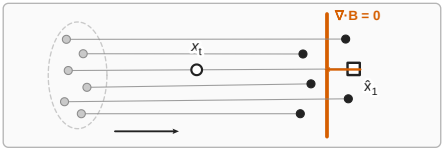
W9: affine term learned next to the network
W10: exact Gaussian field; W11: plus learned affine correction

Shared glyphs



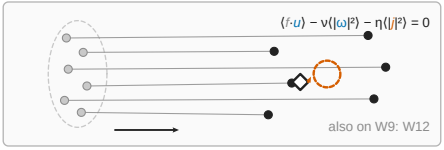
Training loss

W1 Divergence penalty



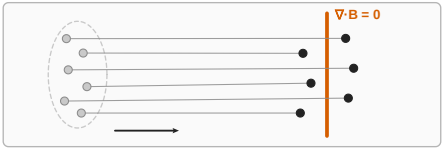
on the end-of-path prediction $\hat{x}_1 = x_t + (1 - t)\, v$

W5 Law-level stationarity loss



constrains a mean over samples, not each sample

W0 Baseline: no constraint



free Fourier components; no mechanism applied

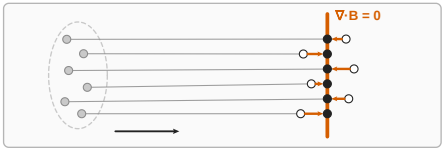
G0, G1, W6 Model families (baselines)



G0: Gaussian per mode; G1: Gaussian with covariance between modes (both no network); W6: diffusion, same network

Sampling procedure

W2 Final Leray projection



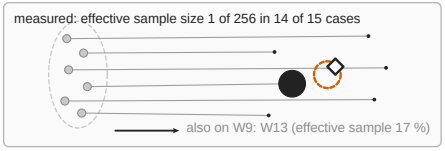
projection of the final samples onto the surface

W3 Projected flow



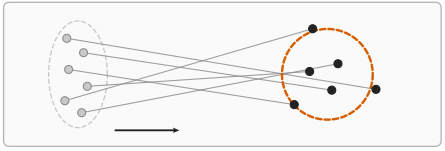
training noise projected + every stage projected

W7 Exponential tilting



reweights the samples on the identities (dot size = weight)

W8 Per-sample rescaling



each sample: small scales rescaled to the energy balance